1. What are the two assumptions that are required in order that the statistic $\frac{\bar{x} - \mu}{s/\sqrt{n}}$ follow a t distribution?

① $X_i \sim N(\mu, \sigma^2)$

② i.i.d

Using the t Table *Find the tabled value of t (t_a) corresponding to a right-tail area a and degrees of freedom given in Exercises 2–6.*

2. $a = .05$, $df = 20$

$t = 1.725$

3. $a = .01$, $df = 18$

$t = 2.552$

Using the t Table II *Find the tabled value of t (t_a) corresponding to a left-tail area a and degrees of freedom given in Exercises 7–11.*

7. $a = .01$, $df = 9$

$t = -2.821$

8. $a = .05$, $df = 8$

$t = -1.860$

17. A random sample of $n = 12$ observations from a normal population produced $\bar{x} = 47.1$ and $s^2 = 4.7$. Test the hypothesis $H_0 : \mu = 48$ against $H_a : \mu \neq 48$ at the 5% level of significance.

$\alpha = 0.05$ $df = n-1 = 12-1 = 11$

(1)Find me the 95% CI for μ

(1)Provide me formula $\bar{x} \pm t_{\frac{\alpha}{2}}$

(2)Calculate the values of the end points of the 95% CI. $\bar{x} \pm t_{\frac{\alpha}{2}} \times \frac{\sqrt{4.7}}{\sqrt{12}}$

$= 47.1 \pm 2.201$

(2)Do the hypothesis test using rejecting region approach

(1) $H_0: \mu = 48$ $H_a: \mu \neq 48$

(2) $\alpha = 0.05$

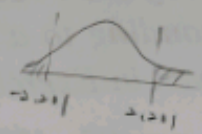
(3) $t_{stat} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t(n-1)$

(4) $t^* = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{47.1 - 48}{4.7/\sqrt{12}} = -1.438$

(5)

(6)

(5)



$t_{11, 0.025} = 2.201$

(6) $T = -1.438 > -2.201$

$\therefore$ non-reject H_0

(3)Do the hypothesis test using the p-value approach

$|t| = 1.44$ $df = 11$

p-value $= 0.18$

Because $0.18 > 0.05$

Fail to reject